

# SMART BUILDING ENERGY INTELLIGENCE

Power BI Portfolio Case Study

From raw building, electricity and weather data to an interactive energy-performance dashboard.

POWER BI • POWER QUERY • DAX • DATA MODELING • ETL

**Portfolio scope:** Power BI only. The public version intentionally excludes Power Apps / Power Automate.

# 1. Business Problem & Objective

The project explores how Business Intelligence can help building and energy managers understand electricity performance across a portfolio of smart buildings.

- Which buildings consume the most electricity?
- When does peak consumption occur?
- How does consumption differ by building type and floor area?
- How does outdoor temperature relate to energy demand?
- Which buildings show unusual consumption behavior compared with their historical baseline?

**Project objective:** transform raw operational data into a reliable, interactive decision-support dashboard with portfolio-level monitoring and building-level diagnostics.

## 2. Data Sources

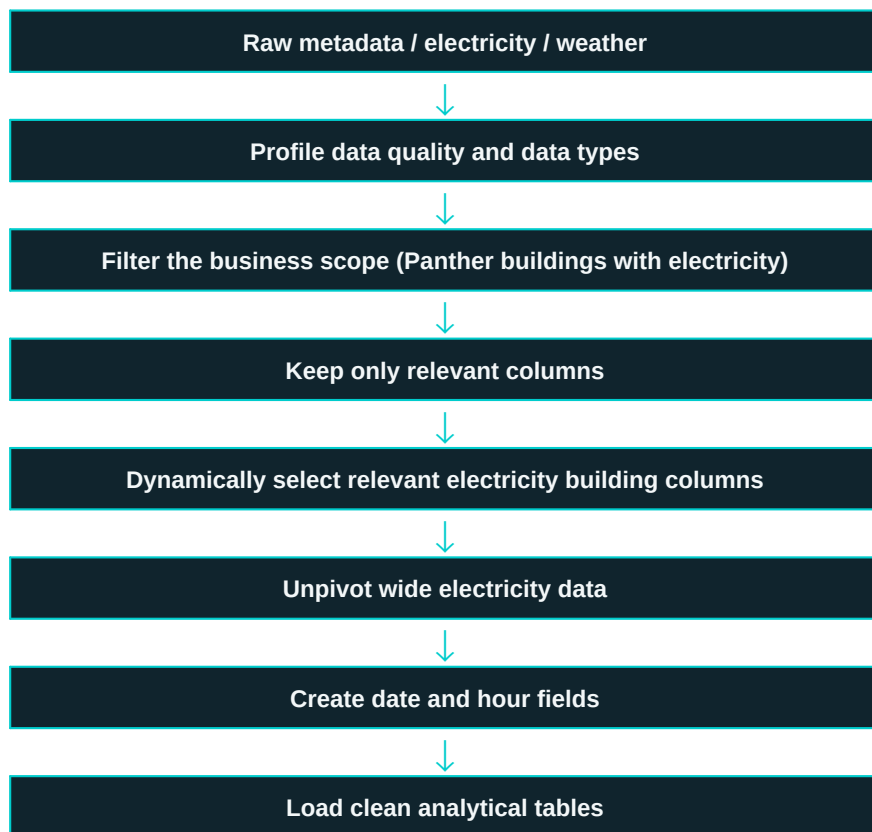
The solution uses three public source groups. The raw data was intentionally not treated as dashboard-ready; data preparation was part of the project.

| Source            | Purpose                         | Examples                                            |
|-------------------|---------------------------------|-----------------------------------------------------|
| Building metadata | Describe each building          | building_id, type, floor area, year built, LEED     |
| Electricity       | Hourly consumption measurements | timestamp + one column per building in the raw file |
| Weather           | External context                | temperature, dew point, precipitation               |

**Why this dataset?** It combines time-series energy data, building characteristics and weather, which makes it suitable for realistic ETL, modeling, performance and diagnostic analysis.

### 3. ETL & Power Query Workflow

Power Query was used to profile, reduce, reshape and prepare the data before modeling.



**Important data-quality rule:** missing values were not automatically replaced with zero. A missing measurement and a genuine zero measurement have different meanings.

## 4. Performance Optimization — Main Technical Challenge

The first implementation unpivoted the full electricity dataset before reducing the building scope. That generated approximately **26.8 million intermediate rows** and made processing unnecessarily expensive.

### Initial approach (inefficient)

~1,000 building columns → Unpivot everything → ~26.8M intermediate rows → filter/join → relevant rows

### Optimized approach

~1,000 building columns → dynamic list of ~105 relevant buildings → select columns first → Unpivot → analytical fact table

**Key learning:** BI performance is not only about DAX. Transformation order matters. Reduce data before expensive transformations whenever possible.

## 5. Dimensional Data Model

The final model separates descriptive dimensions from measurement fact tables.

| Table            | Role                            | Relationship                                     |
|------------------|---------------------------------|--------------------------------------------------|
| Dim_Building     | Building attributes             | 1 → * Fact_electricity via building_id           |
| Dim_Date         | Calendar attributes             | 1 → * Fact_electricity and Fact_weather via date |
| Fact_electricity | Hourly electricity measurements | Filtered by building and date dimensions         |
| Fact_weather     | Weather observations            | Filtered by Dim_Date                             |

**Why this structure?** It avoids repeating building attributes across hourly measurements, keeps relationships clear, and supports reusable DAX measures under different filter contexts.

## 6. DAX & KPI Layer

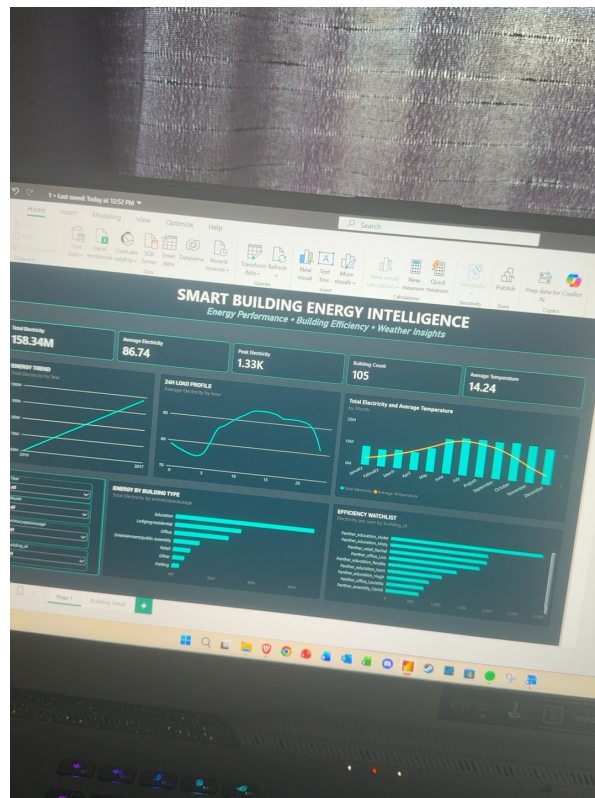
Measures were created for reusable business logic that recalculates dynamically with filters.

| Measure                 | Purpose                                                    |
|-------------------------|------------------------------------------------------------|
| Total Electricity       | Total consumption in the current filter context            |
| Average Electricity     | Average electricity reading                                |
| Peak Electricity        | Maximum recorded electricity value                         |
| Building Count          | Distinct buildings in context                              |
| Average Temperature     | Average external temperature                               |
| Electricity per sqm     | Normalizes consumption by floor area for fairer comparison |
| Expected Electricity    | Historical baseline for a selected building/hour           |
| Deviation from Baseline | Difference between actual behavior and expected baseline   |
| Energy Health           | Rule-based Normal / Warning / Anomaly classification       |

**Note:** electricity values are presented using the source measurement scale. The public portfolio should not label them as kWh, kW or monetary values unless the source documentation explicitly confirms the unit.

## 7. Executive Overview Dashboard

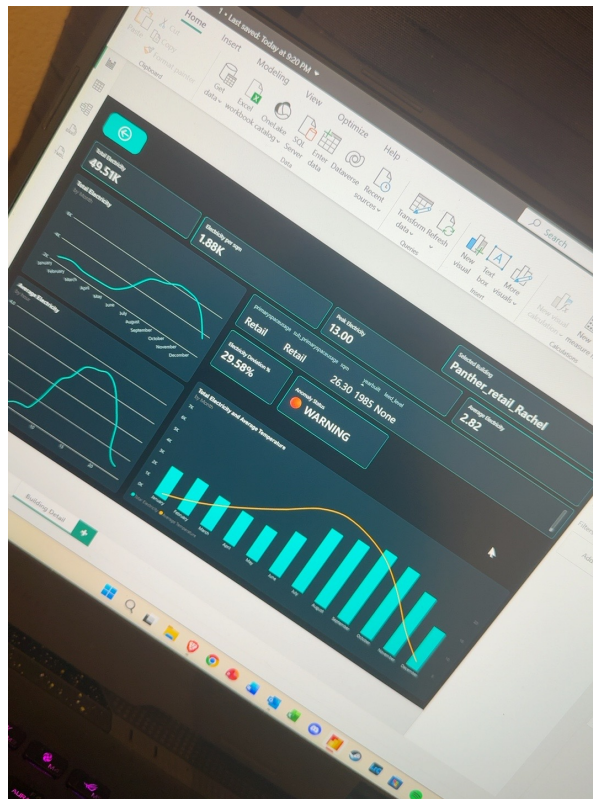
The portfolio page provides interactive filtering by year, month, building type and individual building. It combines high-level KPIs with temporal, operational and comparative analysis.



- Energy Trend — portfolio consumption over time
- 24H Load Profile — when electricity demand occurs
- Energy by Building Type — portfolio composition and comparison
- Energy × Weather — temperature context alongside energy behavior
- Efficiency Watchlist — normalized electricity per square meter

## 8. Building Performance Detail

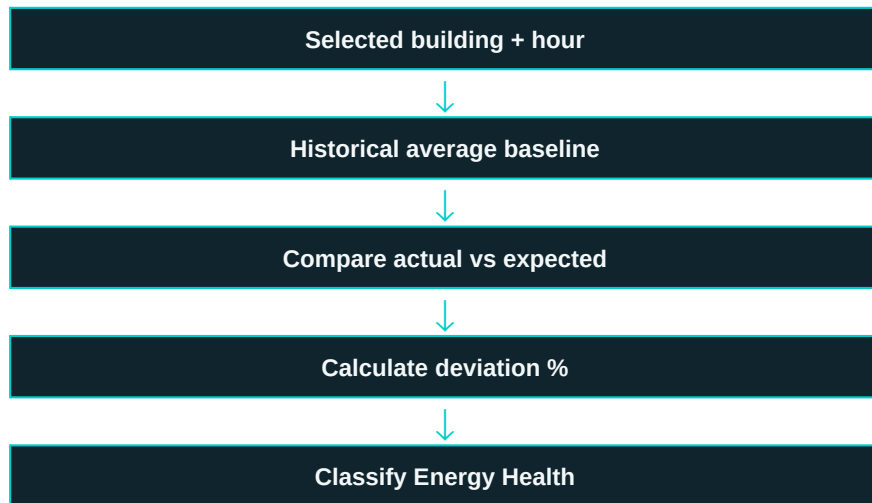
Drill-through allows the user to move from the portfolio view to an individual building without rebuilding the analysis.



- Dynamic selected-building context
- Total, average, peak and normalized electricity KPIs
- Monthly energy profile
- 24-hour load behavior
- Building characteristics (type, area, year, LEED)
- Energy × Weather context
- Energy Health and deviation from baseline

## 9. Energy Health — Baseline Monitoring

The project adds a diagnostic layer beyond descriptive reporting. For a selected building and hour, historical behavior is used to estimate an expected electricity baseline. Actual behavior is compared with that baseline.



**Classification used in the prototype:** Normal for small deviations, Warning for larger deviations, and Anomaly for very large deviations. These are demonstration thresholds and should be calibrated in production.

**Important:** this is baseline/rule-based monitoring, not a trained machine-learning model.

## 10. Key Outcomes & Lessons

The project demonstrates an end-to-end Power BI workflow rather than only dashboard design.

- Business framing before visualization
- Data profiling and careful treatment of missing values
- Efficient Power Query transformation design
- Wide-to-long reshaping with Unpivot
- Dimensional modeling with fact and dimension tables
- Reusable DAX measures and filter context
- Interactive portfolio analysis and drill-through diagnostics
- Performance optimization through early data reduction
- Transparent limitations and technically accurate claims

**Core takeaway:** Good Business Intelligence starts before the dashboard — with understanding the problem, preparing trustworthy data, designing an efficient model, and turning analysis into useful decisions.

## 11. GitHub Repository Structure

Recommended public repository structure:

```
smart-building-energy-intelligence/ ■■■■ README.md ■■■■ dashboard/ ■ ■■■■  
Smart_Building_Energy_Intelligence.pbix ■■■■ screenshots/ ■ ■■■■ executive-overview.png ■ ■■■■  
building-detail.png ■■■■ docs/ ■■■■ Smart_Building_GitHub_Portfolio.pdf
```

**Do not upload:** the huge raw dataset, interview-preparation files, CV/school documents, credentials, tokens, or private company information. Link to the original public dataset in the README instead.

## 12. Portfolio Summary

**Smart Building Energy Intelligence** is an end-to-end Power BI project focused on energy-performance analysis across a portfolio of buildings. It demonstrates Power Query ETL, performance optimization, dimensional data modeling, DAX, interactive dashboard design, drill-through analysis, weather context and baseline-based energy-health monitoring.

**Technology:** Power BI • Power Query • DAX • Dimensional Data Modeling • ETL • Data Visualization

Suggested GitHub repository name: **smart-building-energy-intelligence**